WEEK 2 PROGRESS REPORT

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening | Week Ending: 11th September 2026

Work Completed Against Weekly Objectives

Task	Status	Notes
Model Selection Note	Completed	Selected Google Gemini Flash as the baseline LLM.
Working baseline model interaction	Completed	Integrated Gemini Flash into the application. The baseline sends a single-turn message to the model and returns a response, with no RAG, tools, or conversation memory yet.
Prompt Specification and prompt version history	Completed	Revised from v1.0 (broad "all students" scope) to v1.1 (freshmen-only scope).
10-case prompt evaluation table	Completed	Tested directly in Google AI Studio using the v1.1 prompt; not yet run through the integrated application.

Key Engineering Decisions

- Selected Gemini Flash as the baseline model over Claude Haiku 4.5, GPT-4o-mini, OpenRouter, and Groq (Llama 3.1/3.3). Gemini Flash offers a genuinely free tier requiring no credit card, sufficient capability and context window for grounded Q&A and escalation reasoning, latency and an acceptable privacy trade-off given the project uses only public documents and synthetic data.
- Scoped the baseline model interaction to a single LLM call with no RAG, tools, or memory, matching the constraints documented in Prompt Specification v1.1 – keeping the Week 2 baseline honest about what is actually implemented versus planned for later weeks.
- Separated fee information (in scope) from fee decisions (out of scope) within the prompt's constraints, directly mirroring the AI Boundary Matrix's distinction between what the agent may draft/explain and what requires human approval.
- Revised the prompt from v1.0 (broadly scoped to "students," with an unfounded 150-word cap) to v1.1, correcting the scope to freshmen-only per the Project Charter and fixing the Context section to accurately reflect that no conversation history is passed to the model.

Failures, Challenges and Current Response

- Identity verification gap: the current baseline cannot verify a user's actual year of study, so US-10's non-freshman redirection only triggers when a student explicitly states their status (e.g. "I'm a 3rd year...") or asks something clearly outside onboarding scope. Response: full identity-based detection will require session state, planned for Week 6; the gap is documented in the Prompt Specification rather than left implicit.

Links and Evidence

GitHub repository: https://github.com/SumayyaN/BSE41014_capstone_group_m_eve

ClickUp board: BSE4104_CAPSTONE_PROJECT_BY_GROUP_M_EVE – Team space Week 2

https://app.clickup.com/1200410000000374/v/b/t/1200410000000374

Individual contribution summary

Member	Task Owned	Evidence
Namata Sumayya	Model Selection Note Week 2 progress report	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/requirements/Model%20Selection%20Note%20Group%20M%20Eve.pdf
Mushabe Moses	Working baseline model interaction	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/main/mak-student-support-agent
Aryampa Joshua	Prompt Specification + prompt version history.	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/requirements/Group M - Prompt Specification v1.0.pdf
Nakaye Hildah	10-case prompt evaluation table.	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/evaluation/10_case_Prompt_Evaluation_Table.pdf

Plan for Next Week (Week 3 – Context Engineering and RAG)

- Assemble a controlled corpus (recommended 10-50 documents or equivalent records) and record provenance.
- Implement ingestion, chunking/segmentation, indexing and retrieval.
- Construct model context from retrieved evidence and show sources in the response or trace.
- Create at least 15 RAG test questions: answerable, partially answerable and deliberately unanswerable.
- Document at least three retrieval/grounding failures and what caused them.